

Mapa de Karnaugh - 2

1- a)

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

b)

A \ B C	00	01	11	10
0				
1		1	1	1

c) $F(A, B, C) = AC + AB$

2. $F(A, B, C) = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C$

A \ B C	00	01	11	10
0		1	1	
1	1	1		

$F(A, B, C) = \bar{A}C + A\bar{B}$

3-

AB \ CD	00	01	11	10
00	1			1
01				
11	1	1	1	1
10	1			1

$AB + \bar{B}\bar{D}$

4. a) $(A + \bar{B})(B + C)$

A	B	C	$\bar{B}$	$A + \bar{B}$	$B + C$	$(A + \bar{B})(B + C)$
0	0	0	1	1	0	0
0	0	1	1	1	1	1
0	1	0	0	0	1	0
0	1	1	0	0	1	0
1	0	0	1	1	0	0
1	0	1	1	1	1	1
1	1	0	0	1	1	1
1	1	1	0	1	1	1

ABC	00	01	11	10
0		1		
1		1	1	1

d) $\bar{B}C + AB$

5. a) $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C}$

ABC	00	01	11	10
0	1		1	1
1	1	1		1

c) $\bar{C} + A\bar{B}$

5 AND, 1 OR, 3 NOT $\Rightarrow$ 9 portas
1 AND, 1 OR, 2 NOT $\Rightarrow$ 4 portas

/ /

6- (a)

A\CD	00	01	11	10
00	1			1
01		1	1	
11		1	1	
10	1			1

(c) $B\bar{D} + BD$

7- (a)

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

(d)

A\BC	00	01	11	10
0			1	1
1		1	1	

(e) $(A+B)(B+C)$

8- a)

AB\CD	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	0	1	1	1
10	0	1	1	1

b) $(C+D)(A+\bar{C}+\bar{D})$

c) $\bar{C}D + AC + C\bar{D} = \bar{C}D + C$

d) Resolução de POS, formado pelos 0.

9-

A\BC	00	01	11	10
0	X	1	1	0
1	0	X	1	0

b)

A\BC	00	01	11	10
0	0	1	1	0
1	0	0	1	0

$C(\bar{A}+B)$

c)

A\BC	00	01	11	10
0		1	1	
1		1	1	

C

d) Sem o Don't care, 1 OR, 1 AND, 1 AND $\Rightarrow$ 3 portas
 Com o Don't care, 0 portas, então economizamos 3 portas.